Ashvin Anilkumar

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Summary

Graduate ECE student at UW-Madison specializing in **Robot Learning** with a special interest in **Reinforcement Learning** for Robotic applications with **2 years** of work experience building scalable AI products. Proven experience deploying scalable AI architectures and deterministic C++/ROS2 pipelines for physical robot autonomy.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering — December 2026

- *Relevant Coursework:* Computer Vision, Topics in Advanced Robotics, High-performance Computing, Optimization.

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering — July 2023

- *Additional Coursework:* Minor in Information Technology

Technical Skills

AI/ML Architectures: State Space Models, Transformers, Diffusion Models, Reinforcement Learning

Computer Vision & Perception: 3D Scene Representation, Multi-sensor Fusion, Semantic Segmentation, Pose Estimation

Robotics & Control: Agentic AI Decision-making, ROS 2, Behaviour Trees, Sim-to-Real

Software & Frameworks: PyTorch, Hugging Face, OpenCV, Isaac Sim, MuJoCo.

Databases: Postgres, MySQL, MongoDB, SurrealDB

Programming: C++, CUDA, Python, Rust, JavaScript, Java

Publications

Chrono Agentic: Evidence-Grounded Agents for Executable World Simulation — June 2026

IEEE Access Proceedings (In Review)

Jingquan Wang, Hongyu Wang, Andrew Negrut, **Ashvin Anilkumar**, Radu Serban, Dan Negrut

- Spearheaded the development of *Chrono Agentic*, an inference-time orchestration platform for building executable PyChrono simulations from natural language.

Sample-Efficient On-Robot Reinforcement Learning — Expected Sep 2026

Targeting: IEEE International Conference on Robotics and Automation (ICRA) 2026

Ashvin Anilkumar, Ryan Gao, William Cong, Nicholas Corrado, Josiah Hanna

- Demonstrated sub-20-minute wall-clock policy convergence for continuous control tasks directly on physical hardware, bypassing traditional sim-to-real bottlenecks.

Work Experience

Graduate Research Assistant, University of Wisconsin - Madison, United States — Dec 2025 – Present

Reinforcement Learning | MuJoCo | ROS2 | Simulation | Sim-to-Real | Perception

- Architected sample-efficient reinforcement learning pipelines for continuous control, optimizing policy convergence for physical hardware deployment, advised by **Prof. Josiah Hanna**.
- Engineered deterministic agentic workflows within the PyChrono physics engine to automate complex simulation setups, advised by **Prof. Dan Negrut**.
- Designed and deployed reactive behavior trees on **Booster K1** hardware to execute autonomous, rule-compliant policies of Robocup 2026, to be held in South Korea.

AI Engineer, Backend Development(AI/ML), Qburst Technologies - Trivandrum, India — Aug 2023 – Aug 2025

Scalable AI Search

FastAPI | Langchain | Llamaindex | Postgres | MongoDB | Huggingface

- Developed an AI search bar using **FastAPI** by integrating Retrieval Augmented Generation(RAG) and Natural language to SQL conversion using **Langchain** & **Llamaindex**.
- Deployed a **cloud-native microservices** using FastAPI and Python to handle complex global influence data.
- Built an optimized text embeddings in **Postgres** and hosted Few-Shot Examples in **MongoDB**.
- Implemented **agentic workflows** (RAG), optimizing token usage by **56%** and enhancing system response speed.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.
- Practiced **CI/CD principles** in an Agile environment, collaborating with cross-disciplinary teams to drive software from R&D to production.

AI Report Reviewer System
FastAPI | Langchain | Azure

- Designed the architecture for the project, making the system extremely deterministic and consistent in generating ratings.
- Formulated an AI system for automated evaluation and structured feedback on reports using **Langchain** and **FastAPI**.
- Achieved **89%** alignment with human ratings, improved evaluation accuracy by **35%** and reduced review costs by **67%**.

Research & Academic Projects

On-Robot Learning Apr 2026 – Present
Reinforcement Learning | SERL | HIL-SERL

- Devise a curriculum for direct-on-robot policy training, bypassing sim-to-real transfer limitations on **NAO** humanoids.
- Optimized sample efficiency to train continuous control policies from scratch within a strict **20 mins** wall-clock constraint. Advised by **Prof. Josiah Hanna**.
- Deployed goal-conditioned RL policies for autonomous navigation and dynamic object manipulation.

RoboCup 2026 Dec 2025 – July 2026
Isaac Sim | MuJoCo | ROS2 | Pytorch | Rerun

- Engineered a decentralized multi-agent coordination stack using **ROS 2** and deep reinforcement learning for a fleet of autonomous **Booster K1**s.
- Migrated simulation environments from **Isaac Sim (PhysX)** to **MuJoCo**, enhancing contact dynamics fidelity and boosting training reliability by **40%**.
- Developed the localization algorithm to optimize agent movement in high-density, dynamic environments.
- Bridged the sim-to-real gap by developing a custom deployment pipeline, reducing iterative hardware testing overhead by **20%**.

Performance Analysis of Obstacle Avoidance algorithm Mar 2026 – Apr 2026
CUDA | OpenMP | Perf

- Architected a deterministic obstacle avoidance pipeline utilizing **HOG** feature extraction and SVM classification.
- Profiled throughput and latency bottlenecks, successfully accelerating execution via OpenMP multi-threading and custom CUDA kernels, achieving **8.4x** and **33.5x** speedups from the baseline, respectively.

Real time Visual Odometry using State Space Models Mar 2026 – Apr 2026
State Space Models | Pytorch | ROS2 | Gazebo

- Benchmarked **State Space Models** (SSMs) for real-time visual odometry inference under strict compute constraints within a ROS/Gazebo environment.
- Developed a simulation of a differential drive robot equipped with a monocular RGB camera in Gazebo.
- Synthesized multi-modal datasets of sequential imagery and ground-truth trajectories via precise teleoperation of a differential drive robot.

Learn RL - A Reinforcement Learning Tutorial Jan 2026 – Present
MuJoCo | ROS2 | Pytorch | OpenAI Gym | Stable Baseline 3

- A tutorial hand-tailored by me, which covers basic concepts in the field of reinforcement learning, along with their implementations, encouraging a hands-on way of learning.
- Covers topics like PPO and SAC and uses libraries such as OpenAI Gym, Stable Baseline 3, and physics engines such as Mujoco.

University Rover Challenge: Perception & Autonomous Navigation Sep 2025 – Dec 2025
ROS2 | Gazebo | DepthAI | RViz| RANSAC

- Programmed a real-time **perception and mapping** framework for autonomous navigation on unstructured terrain using **ROS2 Humble**.
- Developed a ground detection algorithm using **RANSAC** and improved the system reliability by **30%**.
- Utilized **DepthAI** and **PyTorch** to deploy semantic segmentation models, enabling the robot to distinguish between navigable terrain and obstacles.
- Implemented **Path Planning** and **Movement** controls using ROS2, integrating stereo vision (DepthAI) and IMU data for sensor fusion.

Teaching Experience

Introduction to Robotics (ME/ECE 439) May 2026 – Present

- Mentored students in the design and deployment of mobile robots and robotic arms with **Prof. Peter Adamczyk** .
- Architected and distributed a custom, optimized ROS image for Raspberry Pi deployment to streamline hardware integration.
- Helped student understand concepts in diverse domains, including kinematics, dynamics, software development, and physical AI.